Cloud LearnX

Mathematics

Circle Theorems — Practice Worksheet

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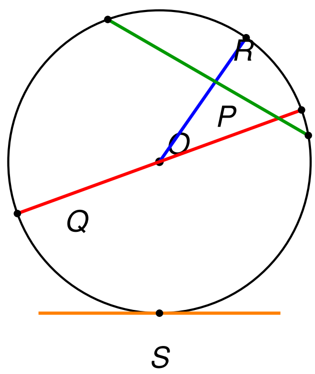
Topic: Circle Theorems

Instructions:

- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

Answer ALL ALL questions.
Write your answers in the spaces provided.
You must write down all the stages in your working.

1 A radio telescope dish used to track spacecraft has a circular support ring, part of which is shown below. Four lines, labelled *P*, *Q*, *R* and *S*, have been drawn onto the diagram by an engineer.



Use the diagram to answer the following.

(a) Write down the letter of the line which is a **tangent** to the circle.

....

(1)

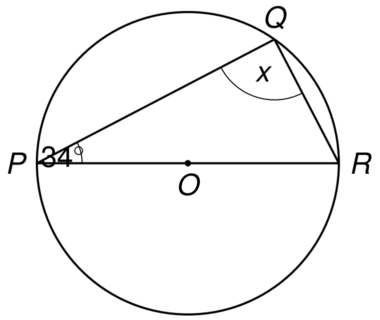
(b) Write down the letter of the line which is a **chord** but **not** a diameter.

....

(1)

(Total for Question 1 is 2 marks)

2 A satellite dish is designed as a circular reflector. Engineers model the circular rim as a circle with centre O. Points P, Q and R lie on the circumference, where PR is a diameter of the circle.



PR is a diameter of the circle.
Angle $QPR = 34°$.

Find the size of angle x.

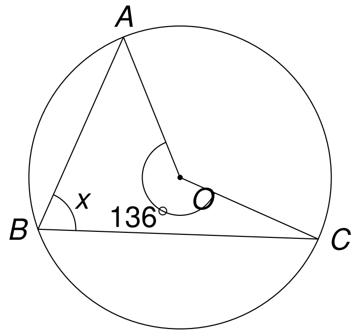
x = ... °

(2)

(Total for Question 2 is 2 marks)

3 A, B and C are points on the circumference of a circle with centre O.

Angle AOC = 136°, where O is the centre of the circle.



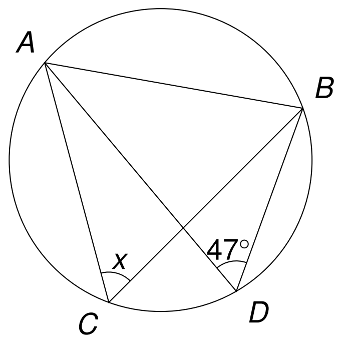
Work out the size of angle x, marked at point B on the circumference.

x = ... °

(2)

(Total for Question 3 is 2 marks)

4 A, B, C and D are points on the circumference of a circle, as shown in the diagram below. Chord AB is subtended by angles at points C and D on the circumference.



Angle $ADB = 47°$.

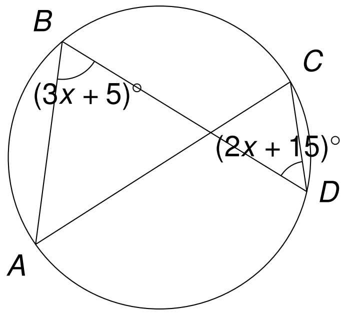
Find the size of angle ACB, marked x on the diagram.

x = ... °

(2)

(Total for Question 4 is 2 marks)

5 *A*, *B*, *C* and *D* are four points on the circumference of a circle, forming a cyclic quadrilateral *ABCD*. This shape is used in the design of a circular stained-glass window depicting a map of the ancient world, on display in a history museum.



Angle *ABD* is $(3x + 5)^\circ$ and angle *BDC* is $(2x + 15)^\circ$.

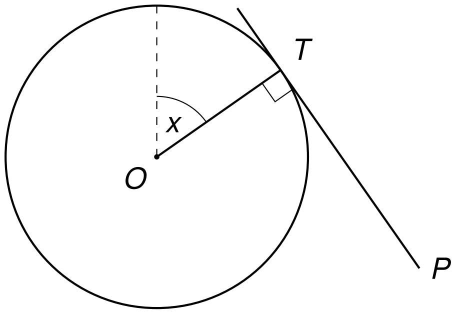
Calculate the size of angle *ABD*.

ABD = ... °

(3)

(Total for Question 5 is 3 marks)

6 A satellite dish is modelled as a circle with centre *O*. A signal travels along a straight line from a point *P* outside the circle, touching the edge of the dish at point *T*, where it forms a tangent to the circle. The straight line *OP* passes through the centre.



OQ is a radius drawn vertically upwards from *O*, and *OT* is another radius drawn to the point of contact *T*, where the angle between them is $x°$.

The line *TP* is a tangent to the circle at *T*, and angle *OTP* = 90° (radius meets tangent).

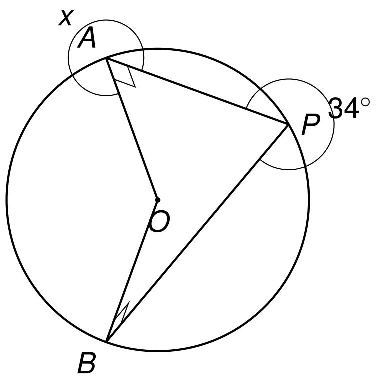
Given that angle *TPO* = 34°, work out the value of *x*.

x = ... °

(3)

(Total for Question 6 is 3 marks)

7 A satellite dish tracking station is represented by point P, outside a circular signal boundary with centre O. Two straight signal paths, PA and PB, are drawn from P so that they just touch the circular boundary at points A and B, as shown in the diagram.



PA and PB are tangents to the circle, and angle $APB = 34°$.

Find the size of angle PAO, marked x on the diagram.

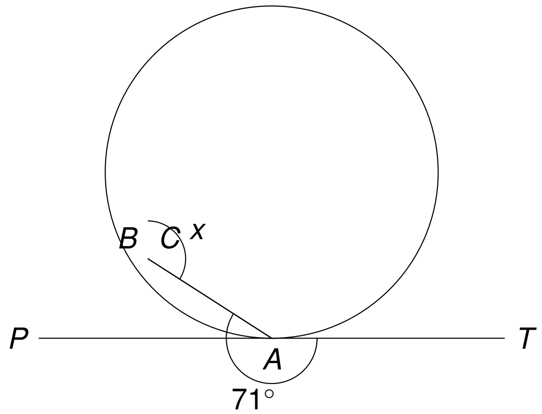
x = ... °

(3)

(Total for Question 7 is 3 marks)

8 A circular observatory dome has a triangular support truss *ABC* fixed to its circular base, as shown in the diagram. *PT* is a tangent to the circle at point *A*, touching the dome's base along the line of a horizon sensor track.

AB and *AC* are chords of the circle. The angle between the tangent *PT* and the chord *AC* is 71°, as shown.



Find the size of angle x, marked at vertex *B*.

Give a reason for each stage of your working.

x = ... °

(3)

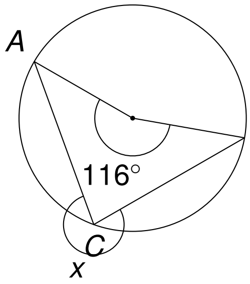
(Total for Question 8 is 3 marks)

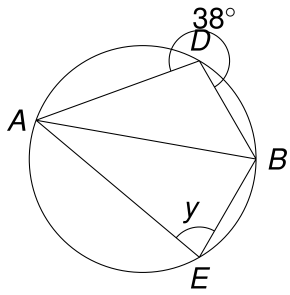
9 A space engineering student is studying satellite dish cross-sections, which are modelled using circles. Four diagrams below each illustrate a different circle theorem.

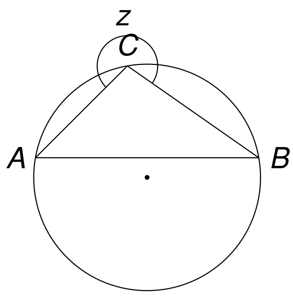
For each diagram:

(i) Name the circle theorem being illustrated.

(ii) Work out the size of the missing angle marked with a letter.







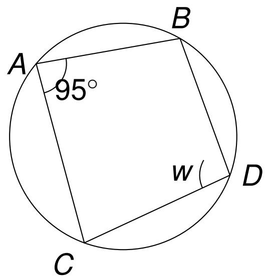
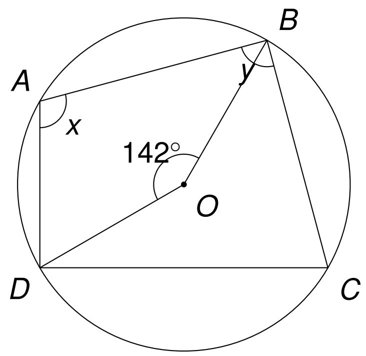


Diagram	Name of theorem	Missing angle
1		x =
2		y =
3		z =
4		w =

(4)

(Total for Question 9 is 4 marks)

10 A, B, C and D are points on the circumference of a circle, centre O. O is the centre of the circle, and BOD is a straight line is **not** the case here — instead O is joined to B and D to form the angle at the centre.



$ABCD$ is a cyclic quadrilateral. The reflex angle BOD is not used; the angle at the centre, angle BOD = 142°, stands on the same arc BD as angle $BAD = x$.

(a) Work out the size of angle x (BAD).

x = ... °

(2)

(b) Calculate the size of angle y (ABC), given that $BCDA$... in fact $ABCD$ is a cyclic quadrilateral, so angle BAD and angle BCD are opposite angles.

Use your answer to part (a) to find angle BCD, then work out the size of angle $y = ABC$, given that angle ADC = 95°.

y = ... °

(2)

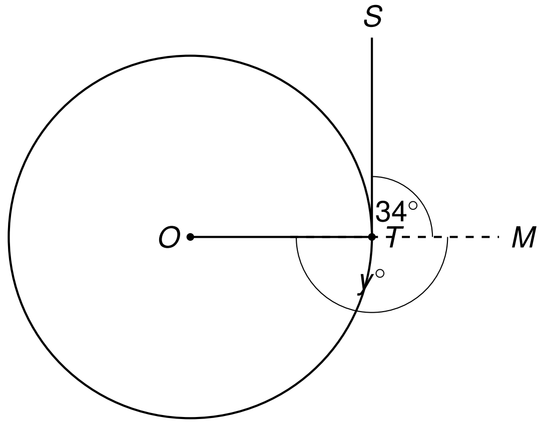
(Total for Question 10 is 4 marks)

11 A satellite dish company is designing a new circular receiver dish, O being the centre of the dish's circular rim.

A straight support strut, ST, is bolted to the rim at point T so that it just touches the rim at T and does not cross into the dish — that is, ST is a tangent to the circular rim at T.

A second support beam runs from the centre O of the dish to the point T, along the radius OT.

The strut ST makes an angle of 34° with a horizontal mounting bar TM, as shown. The mounting bar TM passes through T and is horizontal. The radius OT makes an angle of y° with the horizontal mounting bar TM, on the other side of OT from the strut.



(In the diagram, S is the top end of the strut, O is the centre of the dish, T is the point of contact on the rim, and M is a point on the horizontal mounting bar.)

Work out the value of y, the angle between the radius OT and the mounting bar TM.

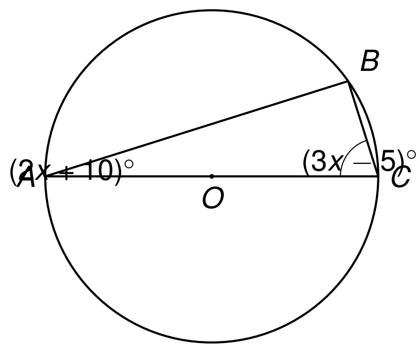
You must show your working.

y = ... °

(3)

(Total for Question 11 is 3 marks)

12 A historian is studying the design of an ancient circular observatory floor, once used to track the rising points of stars on the horizon. Three marker stones A, B and C lie on the circle, with AC passing through the centre O.



AC is a diameter of the circle. Angle $BAC = (2x + 10)^\circ$ and angle $BCA = (3x - 5)^\circ$.

Show that $x = 15$.

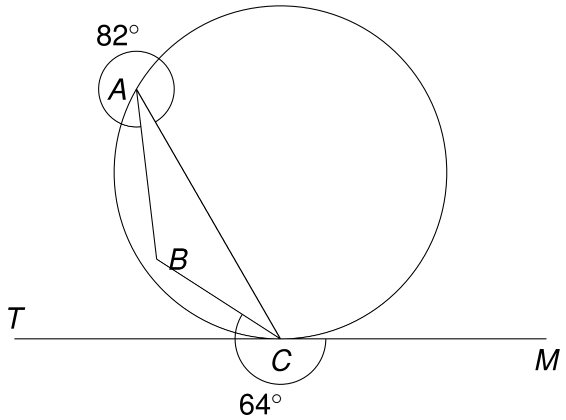
(3)

(Total for Question 12 is 3 marks)

13 A space agency is testing a circular parabolic mirror segment for a new orbital telescope. During calibration, a laser guide beam is fired so that it just grazes the edge of the circular mirror rim at point C, touching it at exactly one point (i.e. the beam is a tangent to the circle at C).

Points A, B and C lie on the circumference of the mirror's circular cross-section. The chord CB is a support strut inside the mirror housing, and chord CA is a second strut. The laser beam TCM is tangent to the circle at C.

The engineers measure the angle between the laser beam and strut CB, on the side nearer M, to be $64°$. They also know that angle $BAC = 82°$.



$\angle BCA$ = ...

(a) Calculate the size of angle BCA.

(2)

$\angle CBA$ = ...

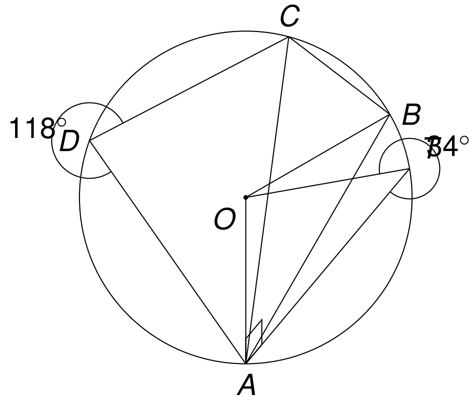
(b) Hence calculate the size of angle CBA, the angle at strut junction B.

(2)

(Total for Question 13 is 4 marks)

14 A space agency is designing a circular observation window for a research module. In the diagram, A, B and D lie on a circle with centre O. TA is a tangent to the circle at A. TBD is not a straight line; instead T is an external point where the tangent at A meets the extended chord OB.

$OA = OB$ (radii of the circle). Angle $OAT = 90°$ (tangent-radius property). Angle $ATB = 34°$. $ABCD$ is a cyclic quadrilateral, with C also on the circle, and angle $ADC = 118°$.



OAT = ... °

(a) Find the size of angle OAT, giving a reason.

(1)

AOB = ... °

(b) Work out the size of angle AOB.

(2)

ABC = ... °

(c) Work out the size of angle ABC.

(1)

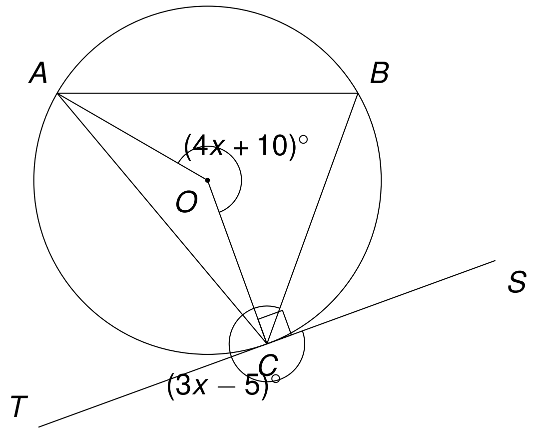
(Total for Question 14 is 4 marks)

15 A space agency is designing a circular observation window for a module. Points A, B and C lie on the circumference of the circular window, with centre O. The line TCS is a tangent to the circle at C.

$OA = OC$ (radii of the circle), and the angle at the centre $AOC = (4x + 10)°$.

The angle between the tangent TC and the chord CB is $BCT = (3x - 5)°$.

B lies on the circle such that angle ABC is the angle in the alternate segment to angle BCT, and angle ABC also lies on the arc such that it is subtended by the same arc AC as the angle at the centre AOC (i.e. ABC is the angle at the circumference standing on arc AC, opposite O).

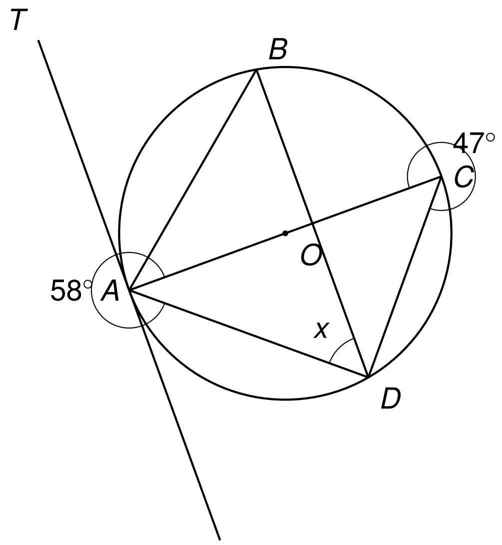


Show that angle ABC and angle BCT are **not** equal unless x takes a specific value, and find the value of x for which angle ABC = angle BCT.

(4)

(Total for Question 15 is 4 marks)

16 An astronomer sketches the circular cross-section of a satellite dish support ring, with O as the centre. TA is a tangent to the circle at A. B, C and D are points on the circumference. AC is a diameter of the circle.



Angle $CAD = 58°$ and angle $DCA = 47°$.

Calculate the size of angle x (= $\angle BDA$), giving reasons for each stage of your working.

x = ... °

(4)

(Total for Question 16 is 4 marks)

TOTAL FOR PAPER IS ?? **MARKS**